

Action with Visual Primitives

Weilong Guo^{*†‡}

Anyverse Dynamics
weilong.guo@anyverse.com

Yuchen Wang^{*}

Tsinghua University
wyuchen0416@zju.edu.cn

Renping Zhou

Tsinghua University
zrp21@mails.tsinghua.edu.cn

Yunfeng Zhang

Anyverse Dynamics
yunshan.zhang@anyverse.com

Rui Fang

Anyverse Dynamics
rui.fang@anyverse.com

Wenda Xu[‡]

Anyverse Dynamics
wenda.xu@anyverse.com

Yuan He

Tsinghua University
heyuan1993@gmail.com

Gao Huang[‡]

Tsinghua University
gaohuang@tsinghua.edu.cn

Abstract: Vision-Language-Action (VLA) models have emerged as a promising paradigm for generalist robotic manipulation. A common design in current architectures maps language instructions and visual observations to actions in a single forward pass. While conceptually simple, this formulation entangles instruction comprehension, spatial scene understanding, and motor control within a single learning objective. As a result, the action expert must implicitly relearn cognitive and perceptual capabilities already present in the pretrained VLM, which can limit both learning efficiency and generalization. We introduce AVP (Action with Visual Primitives), an end-to-end architecture that implements this visual-primitive-centric interface: the VLM infers the next-stage target and emits visual-primitive tokens that condition a flow-matching action expert, with supervision derived from end-effector kinematics. Real-robot experiments on general pick-and-place tasks show that AVP improves the success rate by 27.61% over $\pi_{0.5}$ and outperforms other recent methods, with consistent gains in data efficiency, spatial-compositional generalization, and object-level transfer.

Keywords: Vision-Language-Action, Visual Primitives, Robotic Manipulation, End-to-End

1 Introduction

Building general-purpose robots that perform diverse manipulation tasks in the open world is a long-standing goal of embodied AI. Inspired by the success of large pre-trained models in language and vision, recent research has introduced Vision-Language-Action (VLA) models [1, 2, 3, 4, 5], which fine-tune Vision-Language Models on large-scale robotic datasets [6, 7, 8, 9] to unify language understanding, visual perception, vision-language alignment, and action generation in a single framework. Pioneering works such as RT-2 [10], OpenVLA [1], and $\pi_{0.5}$ [3] have demonstrated this paradigm’s effectiveness across a variety of complex multi-task scenarios.

^{*}Equal contribution.

[†]Project Leader.

[‡]Corresponding author.

The domino moves from (2,2) to (8,4) in vertical posture.	Observation for Action Expert				
	Subtask Instruction	"Grab the domino at position (2,2) with the right arm."			"Take the domino from the right arm with the left arm and place it vertically at position (8,4)."
	Visual Primitive	Pose primitives			

Put the white mug on the left plate and put the yellow and white mug on the right plate.	Observation for Action Expert					
	Subtask Instruction	"Approach the white mug to grasp it."	"Carry the white mug to the right plate and release it."		"Approach the yellow mug to grasp it."	"Carry the yellow mug to the left plate and release it."
	Visual Primitive	Goal primitives				

Use the right arm to move the red soldier from (7,0) to (1,3).	Observation for Action Expert					
	Subtask Instruction	"Grab the red soldier at position (7,0) with your right arm."			"Move and place red soldier at position (1,3)."	
	Visual Primitive	Memory primitives				

Press the buttons in order.	Observation for Action Expert					
	Subtask Instruction	"Press yellow button."	"Press pink button."	"Press green button."		
	Visual Primitive	Order and Memory primitives				

Figure 1: **Different visual primitives of our AVP.**

However, existing VLA architectures have yet to fully leverage the semantic and spatial reasoning capabilities that VLMs acquire through large-scale pre-training. A central challenge is that current designs require the action expert to map raw VLM features directly to motor commands, mixing perception, reasoning, and control within a single learning objective [11, 12, 13]. This coupling makes it hard for the action expert to separate transferable motor skills from task-specific visual context, which often shows up as weak generalization when objects, layouts, or environments change at test time. Several lines of work seek to address this by introducing explicit intermediate representations. (i) Planning-centric approaches [3, 14, 15] such as $\pi_{0.5}$ break instructions into *subtask descriptions*, letting the action expert focus on atomic motor skills and thereby improving compositional generalization. Still, the interface remains purely linguistic and can struggle to convey fine-grained spatial distinctions, especially in visually dense scenes or among geometrically similar entities. (ii) $\pi_{0.7}$ [16] and world-action models [17, 18] provide *subgoal images* or predicted

future frames as a visual target for the action expert. While spatially informative, dense pixel predictions still require the action expert to re-discover what in the scene actually matters for the task. (iii) Visual-prompt methods such as Point-VLA [19], VP-VLA [20] and TraceVLA [21] turn VLM outputs into spatial markers drawn on the input image, offering direct spatial grounding. However, these approaches rely on external models to produce the prompts, leading to multi-stage pipelines that add inference latency and risk passing errors from one stage to the next.

This motivates us to revisit a fundamental question: *how should responsibilities be partitioned between the VLM and the action expert?* Ideally, the two should function as a cohesive whole yet maintain a clear division of learning responsibilities—the VLM reasons about what to do and where, while the action expert focuses solely on how to execute. Building on these observations, we propose **AVP (Action with Visual Primitives)**, an end-to-end VLA architecture that places visual communication at the center of the VLM $\leftrightarrow$ action-expert boundary. AVP explicitly demarcates the learning responsibilities of the two modules through a *visual-primitive-centric communication protocol*: the VLM emits visual primitives—points, bounding boxes, or any of the forms exemplified in fig. 2—that the action expert reads as direct, visually grounded instructions to execute. Our AVP consists of three components: a pretrained VLM, an autoregressive decoder, and an action expert. Given a language instruction and multi-view observations of the robot, the VLM parses the current subtask and predicts both the next-stage target and the visual primitives used to represent it. This latent prediction is then projected from the language space into the visual space, yielding image tokens overlaid with visual primitives. These tokens, in turn, serve as the condition that drives the action expert to produce robot actions. Such a design is both simple and expressive. It offloads complex spatial reasoning from the action expert, implicitly encodes world-model-like predictions of future states, and extends naturally to memory-augmented variants.

Through real-world robot experiments, we demonstrate that AVP achieves strong manipulation performance across multiple tasks, including Chinese chess manipulation, domino placement, and general object pick-and-place. Beyond standard task execution, AVP exhibits strong spatial compositional generalization, enabling direct execution of unseen transitions, as well as effective zero-shot generalization to unseen objects. Under identical experimental conditions, our method improves the overall manipulation success rate by 27.61% over the $\pi_{0.5}$ baseline.

In summary, our contributions are as follows:

- We revisit the division of responsibilities between the VLM and the action expert, and propose visual primitives—spatially grounded tokens produced internally by the VLM—as an explicit communication medium that replaces the implicit, unstructured boundary in current VLAs.
- We introduce AVP, which implements this idea end-to-end: the VLM infers the next-stage target and emits visual-primitive tokens that condition a flow-matching action expert, with supervision derived from end-effector kinematics.
- Experiments on multiple real-robot pick-and-place tasks show a 27.61% success-rate improvement over $\pi_{0.5}$, along with meaningful gains in spatial-compositional and object-level generalization.

2 Related Work

Vision-Language-Action Models. Vision-Language-Action (VLA) models unify visual, language, and action modalities into a single policy, emerging as a practical paradigm for robotic control [22, 23, 4, 2, 24, 5]. By leveraging the inherent comprehension capabilities of Large Language Models (LLMs) and Vision-Language Models (VLMs), and pre-training on large-scale, heterogeneous multimodal datasets, current VLAs demonstrate strong transferability and generalization across diverse tasks. Previous works such as RT-2 [10], and OpenVLA [1] have explored the feasibility and scaling behavior of training on large scale robot datasets. Recent advancements, such as $\pi_{0.5}$ [3] and GR-3 [25], have further enhanced the robustness of action policies and real-world deployment efficacy by incorporating flow-matching action experts [26, 27]. Building upon this line of

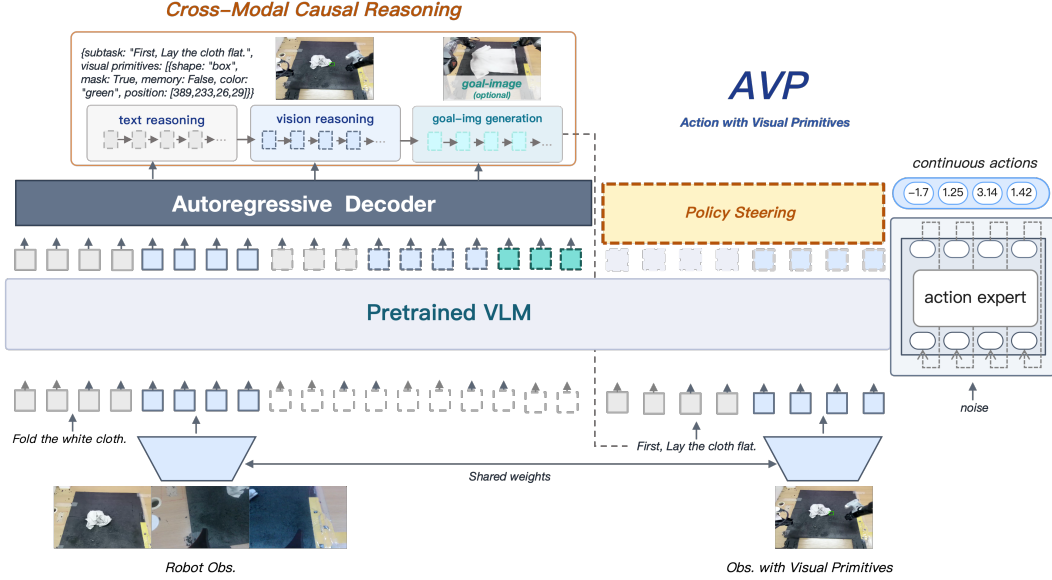


Figure 2: Overview of the AVP framework.

work, cutting-edge generalist architectures like DM0 [28] and LDA [29] push the limits of physical AI through embodied-native training recipes or joint multi-objective latent dynamics optimization. However, although existing VLAs exhibit basic spatial understanding and grounding capabilities derived from their VLM backbones, they often lack sufficient spatial reasoning proficiency when guided by complex natural language instructions [30, 31, 32]. Consequently, a significant gap persists between high-level language directives and the precise low-level visual observations required for fine-grained execution.

Visual Prompting as Intermediate Representations for VLAs. Intermediate visual representations play a crucial role in guiding robotic behavior and facilitating complex control tasks. Existing methods usually employ auxiliary pre-trained vision models—such as the Segment Anything Model (SAM) [33, 34] or generic VLMs [35, 36, 37]—to ground natural language instructions into explicit visual cues directly on the input image, utilizing formats like key points [38, 39, 40], bounding boxes [20, 19, 41], masks [42, 43] and trajectories [44, 45, 21]. Although such structured spatial prompting enhances spatial awareness and instruction-following capabilities in complex environments, this cascaded, dual-system architecture inherently introduces compounded precision errors and significant execution latency. Furthermore, the generation of these visual prompts often relies on task-specific, heuristic designs, which severely limits their broad generalizability. Our approach overcomes these critical limitations by modeling visually prompted images at the discrete visual token level. Through an end-to-end co-optimization framework, we successfully internalize the spatial reasoning capabilities of pre-trained vision models directly into the VLA policy, entirely bypassing the overhead and fragility of external modular dependencies.

3 Method

3.1 Preliminaries: VLA Policies and Visual-Primitive Interfaces

In a typical Vision-Language-Action (VLA) framework, robotic manipulation is formulated as language-conditioned action prediction from visual observations and robot states. At time step t , the policy observes the current multi-view image observation o_t , a language instruction l , and the robot proprioceptive state s_t (e.g., joint angles or end-effector pose), and predicts a horizon of future actions $a_{t:t+h}$. Existing VLAs [4, 3, 25] typically parameterize this mapping as an action policy π_θ :

$$a_{t:t+h} = \pi_\theta(o_t, l, s_t), \quad (1)$$

where h denotes the action horizon.

While this direct formulation is effective for many language-conditioned manipulation tasks, it provides only an implicit interface between the VLM representation and the action expert. Task-relevant spatial cues remain entangled with global visual-language features, rather than being exposed as explicit guidance for action generation. As a result, the action expert must learn semantic parsing, spatial localization, and motion generation jointly from demonstrations, which can reduce data efficiency and weaken generalization under object or layout shifts.

Recent works such as VP-VLA [20] and PointVLA [19] suggest that spatially explicit visual cues can provide a useful interface between high-level task understanding and action generation. We view these methods as instantiating an explicit visual interface in two stages: a task-conditioned visual module first predicts a spatially grounded visual primitive, which is then injected back into the observation as an explicit visual augmentation. A VLA model subsequently predicts actions conditioned on this augmented observation. Formally,

$$p_t = \mathcal{M}_{vp}(o_t, l), \quad \tilde{o}_t = \text{Overlay}(o_t, p_t), \quad (2)$$

where p_t denotes a task-relevant visual primitive, such as a point, bounding box, or other compact spatial mark, and $\tilde{o}_t$ denotes the visually augmented observation. The resulting augmented input is then used for action prediction:

$$a_{t:t+h} = \pi_\theta(\tilde{o}_t, l, s_t). \quad (3)$$

3.2 AVP: Action with Visual Primitives

A straightforward way to instantiate the above interface is to use external vision foundation models, such as SAM [33] or Grounding DINO [46], to generate visual primitives that are overlaid on the raw observation o_t . Methods such as VP-VLA and Point-VLA can be viewed as following this strategy. While effective, this cascaded design introduces an additional perception stage before action generation, which increases inference overhead and makes the overall system sensitive to upstream localization errors. In fine-grained manipulation tasks such as Chinese chess, even small prompt localization drift can alter the intended grasp target or placement cell. In addition, the prompting stage is separated from the action model, limiting joint end-to-end optimization.

To address these limitations, AVP internalizes the visual-primitive interface within a single end-to-end VLA architecture. By reusing the VLM and visual encoder already present in the VLA, AVP predicts task-relevant visual primitives without introducing an external perception module, and enables primitive generation and action prediction to be optimized jointly.

Given the instruction l and observation o_t , the VLM produces multimodal context tokens, from which the visual-primitive decoder autoregressively predicts a discretized visual primitive for the next stage of execution:

$$p_t = D_\psi(\text{VLM}(o_t, l)). \quad (4)$$

The predicted primitive p_t encodes the next-stage subtask together with its associated spatial target, and serves as the explicit condition for subsequent action prediction. To make this interface directly consumable by the action expert, we project the predicted primitive into the visual token space:

$$z_t^{vp} = \text{Proj}(p_t, o_t), \quad (5)$$

where z_t^{vp} denotes the projected visual-primitive tokens. These tokens are then fused with the original multimodal tokens to form an augmented multimodal representation z_t^{aug} , which is used by the action expert for action prediction:

$$a_{t:t+h} = \pi_\theta(z_t^{aug}, s_t). \quad (6)$$

Through this design, the visual primitive acts as an explicit communication channel between the VLM and the action expert: the VLM specifies the task-relevant target in the visual space, while the action expert focuses on motion execution.

We supervise the visual-primitive decoder with an auxiliary visual-primitive prediction loss:

$$\mathcal{L}_{vp} = \mathcal{L}_{CE}(p_t, y_t^{vp}), \quad (7)$$

where y_t^{vp} denotes the ground-truth visual primitive derived from end-effector kinematics. The action expert is trained with the standard action prediction objective $\mathcal{L}_{act}$, and the overall training objective is

$$\mathcal{L} = \mathcal{L}_{act} + \lambda \mathcal{L}_{vp}, \quad (8)$$

where λ balances action learning and primitive supervision.

At inference time, AVP requires only the raw observation, language instruction, and robot state. Visual primitives are generated internally without any external detector, segmenter, or manually overlaid prompt.

4 Experiments

In this section, we evaluate AVP on a suite of real-world robotic manipulation tasks. We first describe the experimental setup and task configurations in [section 4.1](#). We then present quantitative results on multiple manipulation benchmarks in [section 4.2](#). Finally, in [section 4.3](#), we analyze the generalization capability of the proposed method and study the effectiveness of visual prompting.

4.1 Experimental Setup

Task Description. We evaluate our method on three real-world manipulation tasks, including Chinese chess manipulation, domino placement, and general object pick-and-place, as illustrated in [fig. 2](#). The Chinese chess manipulation task serves as the primary benchmark for evaluating spatially precise sequential manipulation. The chessboard contains densely arranged and visually similar chess pieces, posing significant challenges for fine-grained spatial reasoning and manipulation accuracy. The domino placement task further evaluates high-precision bimanual manipulation with orientation constraints, requiring accurate target orientation alignment in addition to positional placement. Finally, the general object pick-and-place task evaluates manipulation generalization across objects with diverse visual appearances and geometric shapes.

Dataset, Training, and Evaluation Metrics. We train all policies using real-world expert demonstrations, including 11.2 hours for Chinese chess manipulation, 1.7 hours for domino placement, and 1.7 hours for general object pick-and-place. All policies are trained with a batch size of 64; the Chinese chess policy is trained for 40k steps, while the domino and general object policies are trained for 10k steps. Unless otherwise specified, we report success rates for *Instruction Following*, *Pick Success*, and *Place Success*. Here, *Instruction Following* evaluates whether the policy selects the correct manipulator according to the input instruction. For the domino placement task, we additionally report *Orientation Success* due to its target pose constraints.

Baseline and Robot Embodiment. We build our method on top of the $\pi_{0.5}$ framework and use it as the primary baseline for comparison. Experiments are conducted on an AgileX Piper tabletop platform consisting of two 6-DoF manipulators with parallel grippers. This dual-arm setup defines a unified 14-dimensional action space used for both training and evaluation. For fair comparison, baseline methods are reproduced under the same robot embodiment, training data, and evaluation protocol whenever applicable. For methods that do not originally support bimanual control, we adapt their action heads to this unified action space while preserving the original model architecture and training protocol as much as possible.

4.2 Main Evaluation

In this section, we evaluate the proposed method across three real-world manipulation tasks and compare it against baseline approaches under a unified evaluation protocol.

Table 1: Quantitative evaluation on the Chinese chess manipulation task. All metrics are reported as success rates (%). Ave. denotes the average success rate across all evaluation metrics, and Latency denotes the average inference time per instruction.

Method	Instr.	Pick	Place	Ave.	Latency	Method	Instr.	Pick	Place	Ave.	Latency
$\pi_{0.5}$	74.00	72.00	42.00	62.67	0.16 s	DM0	50.00	30.00	15.00	31.67	0.52 s
Point-VLA (Qwen)	—	—	—	—	—	LDA	95.00	20.00	10.00	41.67	—
Point-VLA (Kimi)	65.28	47.22	31.94	48.15	37.32 s	π_0	62.50	45.83	25.00	44.44	0.16 s
Ours	98.61	90.28	81.94	90.28	0.27 s	StarVLA	—	—	—	—	—

Method	Pick	Place	Orien.	Ave.
$\pi_{0.5}$	87.50	64.58	93.75	81.94
Ours	100.00	64.58	100.00	88.19
Imp.	+12.50	0.00	+6.25	+6.25

Table 2: **Domino placement results.** Metrics are success rates (%).

Method	Instr.	Pick	Place	Ave.
$\pi_{0.5}$	100.00	71.79	23.08	64.96
Ours	100.00	90.24	68.29	86.18
Imp.	0.00	+18.45	+45.21	+21.22

Table 3: **General object manipulation results.** Metrics are success rates (%).

Experiment I: Chinese chess manipulation. We evaluate the system using a predefined game record. To isolate spatial manipulation capabilities, we exclude capture moves; thus, each action simply requires moving a piece from its initial position to a designated target. A move is successful if the piece accurately covers the target board intersection.

To evaluate LLM-based annotation, we adapt the Point-VLA [19] framework as an additional baseline. While the original Point-VLA utilizes Claude and Gemini, we instantiate it using Qwen3.6-Plus and Kimi-K2.5, accessed via Alibaba Cloud Bailian APIs. Due to the high inference latency of these commercial APIs, we evaluate these Point-VLA variants on a subset of 20 instructions, whereas all other methods are tested on the full 72-instruction sequence.

As shown in Table 1, our method consistently outperforms all baseline approaches across all evaluation metrics, including Instruction Following, Pick Success, and Place Success. In addition, our method demonstrates substantially lower inference latency compared to large-language-model-based annotation approaches that rely on online API calls, highlighting its advantage in execution efficiency and real-time deployment.

Experiment II: Domino placement. This task requires the robot to place a domino at a target position with a specified horizontal or vertical orientation. We evaluate the system on 48 instructions against the $\pi_{0.5}$ baseline. A trial is successful if the final position error is within one domino thickness and the angular deviation from the target orientation is under 10° .

As shown in Table 2, the proposed method improves both pick and orientation success rates over the $\pi_{0.5}$ baseline while maintaining comparable placement accuracy. In contrast to the baseline policy, which frequently exhibits repeated grasping and grasp failure behaviors, our method achieves more stable manipulation execution.

Experiment III: General object pick-and-place. This experiment evaluates the proposed method on general object manipulation across diverse object appearances and shapes. As shown in Table 3, the proposed method substantially improves both grasping and placement success rates over the $\pi_{0.5}$ baseline while maintaining perfect correct-manipulator accuracy.

Across all three tasks, the consistent improvements validate the versatility of our approach. We attribute these gains to the proposed decoupling between understanding and action. By introducing

Method	Instr.	Pick	Place	Ave.
$\pi_{0.5}$	0	0	0	0
Ours	100	90	60	83

(a) Spatial compositional generalization on unseen direct transitions. Metrics are reported as success rates (%).

Setting	$\pi_{0.5}$	Ours
Chessboard + Direct	0/8	8/8
Chessboard + Indirect	7/8	8/8
White cloth + Direct	0/8	7/8
White cloth + Indirect	0/8	8/8

(b) Cross-domain generalization on representative unseen objects.

Table 4: **Generalization results.** Left: spatial compositional generalization where the model is evaluated on unseen direct board-to-board transitions. Right: zero-shot cross-domain generalization to unseen objects under different background conditions and instruction protocols.

structured visual prompts as intermediate spatial representations, the policy is relieved from implicitly learning complex end-to-end instruction-to-action mappings, leading to improved robustness and execution stability in diverse real-world settings.

4.3 Analysis

Our method explicitly decouples *understanding* and *action*, enabling the policy to execute actions based on visual prompts while reducing reliance on fixed instruction-action correspondences. Under this formulation, the model first grounds the task in the visual space and then performs corresponding actions. To validate this property, we conduct generalization studies on the Chinese chess manipulation task from two perspectives: (i) spatial compositional generalization, and (ii) cross-domain generalization to unseen objects.

Spatial Generalization. In the training data, all trajectories are decomposed through predefined board-external intermediate waypoints, such that the model only observes two-stage transitions ($A \rightarrow C \rightarrow B$) and never direct board-to-board transitions ($A \rightarrow B$). We train the model on 12.8 hours of demonstrations for 50k steps and evaluate it on 50 unseen instructions requiring direct point-to-point execution. This setting evaluates whether the model can compose learned sub-trajectories into novel actions. As shown in Table 4a, the $\pi_{0.5}$ baseline fails to generalize to unseen direct transitions and cannot produce valid executable actions. In contrast, our method achieves strong compositional generalization, suggesting that the policy can recombine learned motion patterns into novel behaviors beyond the transition structures observed during training. We attribute this capability to visual prompting, which enables the policy to ground language instructions in spatial representations.

Cross-Domain Generalization. We further evaluate whether a policy trained solely on Chinese chess manipulation data can generalize to unseen objects in a zero-shot manner. As illustrated in appendix fig. 3, we evaluate zero-shot generalization on a diverse set of unseen objects spanning multiple categories. We additionally evaluate under different background conditions and instruction variations. The “Indirect” setting follows the same decomposed transition protocol used during training.

As shown in Table 4b, our method achieves consistently high success rates across unseen objects, background conditions, and instruction variations, despite being trained solely on Chinese chess manipulation data. In contrast, the $\pi_{0.5}$ baseline degrades substantially under distribution shifts, particularly under the “Direct” setting. While its performance partially recovers under the “Indirect” setting, the baseline remains sensitive to changes in object appearance and environmental conditions. These results suggest that visual prompting improves the transferability of spatial manipulation behaviors beyond the original training domain. Additional qualitative examples are shown in appendix fig. 4.

Visual Primitive Ablation. We conduct an ablation study on the Chinese chess manipulation task to evaluate the effect of different visual primitive components, including box-based spatial highlighting, background masking, and memory-enhanced prompts. As shown in table 5, box prompts improve manipulation performance, and adding background masking provides additional gains. Incorporating memory achieves the best overall performance, suggesting that the proposed visual prompt-

Visual Primitive	Instr.	Pick	Place	Ave.
None	100	70	64	78
Box	100	82	68	83
Box + Mask	100	86	70	85
Box + Mask + Mem.	100	94	78	91

Table 5: **Effect of visual primitive design.** Metrics are reported as success rates (%).

ing framework can naturally incorporate additional spatial context across manipulation stages. More ablations on visual prompt design are provided in [section A.1](#).

Scalability. Beyond the benchmarks above, we view the visual-primitive interface as offering a path toward scaling AVP along two complementary axes. *(i) Low-cost training data.* Primitive supervision is derived directly from the end-effector kinematics that every robot already records as part of its proprioceptive state. To deploy AVP on a new platform, the only additional step is a one-time camera calibration that fixes the 3D-to-2D projection; from that point on, every demonstration trajectory automatically yields paired primitive labels at essentially zero additional annotation cost. This avoids the per-sample human labeling required by external-detector-based approaches and keeps data collection close in cost to that of standard imitation-learning pipelines. *(ii) Compact form, easy to extend along time.* Visual primitives are intentionally lightweight—a point, a bounding box, or a small set of them per frame—and the same form extends cleanly along the temporal axis: adding memory primitives that persist across frames, or order primitives that index a sequence of targets (cf. [fig. 1](#)), requires no change to the policy architecture or to the supervision pipeline. This keeps the marginal cost of supporting longer-horizon or memory-dependent tasks low and allows the same model to cover an increasing range of task structures. Together, these properties suggest that AVP can be scaled beyond the in-house benchmarks reported here while keeping per-task cost effectively bounded; a systematic study of this scaling behavior is left to future work.

5 Conclusion

We presented AVP, an end-to-end VLA architecture that introduces visual primitives as an explicit communication interface between the VLM and the action expert. By decoupling target inference from motion execution, AVP relieves the action expert from implicitly learning spatial grounding from complex multimodal inputs. Crucially, our supervision pipeline derives primitive labels directly from end-effector kinematics, eliminating manual annotation or external perception modules. Real-world experiments across three manipulation tasks demonstrate that AVP consistently outperforms the $\pi_{0.5}$ baseline, achieving an overall success-rate gain of 27.61% alongside strong spatial-compositional generalization.

Despite these advantages, several limitations remain. The sequential, two-stage inference (primitive prediction followed by action generation) introduces additional runtime latency compared to single-pass policies. Furthermore, the supervision pipeline relies on precise hand-eye calibration, which makes the system sensitive to camera extrinsic drifts or physical disturbances.

Acknowledgments

We would like to express our gratitude to Anyverse Dynamics for providing the computing resources, robotic hardware platforms, and technical support that made this work possible.

References

- [1] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- [2] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [3] P. Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- [4] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu. Rdt-1b: a diffusion foundation model for bimanual manipulation. *arXiv preprint arXiv:2410.07864*, 2024.
- [5] O. M. Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, et al. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024.
- [6] O. X.-E. Collaboration, A. O’Neill, A. Rehman, A. Gupta, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain, A. Tung, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Gupta, A. Wang, A. Kolobov, A. Singh, A. Garg, A. Kembhavi, A. Xie, A. Brohan, A. Raffin, A. Sharma, A. Yavary, A. Jain, A. Balakrishna, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Wulfe, B. Ichter, C. Lu, C. Xu, C. Le, C. Finn, C. Wang, C. Xu, C. Chi, C. Huang, C. Chan, C. Agia, C. Pan, C. Fu, C. Devin, D. Xu, D. Morton, D. Driess, D. Chen, D. Pathak, D. Shah, D. Büchler, D. Jayaraman, D. Kalashnikov, D. Sadigh, E. Johns, E. Foster, F. Liu, F. Ceola, F. Xia, F. Zhao, F. V. Frujeri, F. Stulp, G. Zhou, G. S. Sukhatme, G. Salhotra, G. Yan, G. Feng, G. Schiavi, G. Berseth, G. Kahn, G. Yang, G. Wang, H. Su, H.-S. Fang, H. Shi, H. Bao, H. B. Amor, H. I. Christensen, H. Furuta, H. Bharadhwaj, H. Walke, H. Fang, H. Ha, I. Mordatch, I. Radosavovic, I. Leal, J. Liang, J. Abou-Chakra, J. Kim, J. Drake, J. Peters, J. Schneider, J. Hsu, J. Vakil, J. Bohg, J. Bingham, J. Wu, J. Gao, J. Hu, J. Wu, J. Wu, J. Sun, J. Luo, J. Gu, J. Tan, J. Oh, J. Wu, J. Lu, J. Yang, J. Malik, J. Silvério, J. Hejna, J. Booher, J. Tompson, J. Yang, J. Salvador, J. J. Lim, J. Han, K. Wang, K. Rao, K. Pertsch, K. Hausman, K. Go, K. Gopalakrishnan, K. Goldberg, K. Byrne, K. Oslund, K. Kawaharazuka, K. Black, K. Lin, K. Zhang, K. Ehsani, K. Lekkala, K. Ellis, K. Rana, K. Srinivasan, K. Fang, K. P. Singh, K.-H. Zeng, K. Hatch, K. Hsu, L. Itti, L. Y. Chen, L. Pinto, L. Fei-Fei, L. Tan, L. J. Fan, L. Ott, L. Lee, L. Weihs, M. Chen, M. Lepert, M. Memmel, M. Tomizuka, M. Itkina, M. G. Castro, M. Spero, M. Du, M. Ahn, M. C. Yip, M. Zhang, M. Ding, M. Heo, M. K. Srirama, M. Sharma, M. J. Kim, M. Z. Irshad, N. Kanazawa, N. Hansen, N. Heess, N. J. Joshi, N. Suenderhauf, N. Liu, N. D. Palo, N. M. M. Shafiullah, O. Mees, O. Kroemer, O. Bastani, P. R. Sanketi, P. T. Miller, P. Yin, P. Wohlhart, P. Xu, P. D. Fagan, P. Mitrano, P. Sermanet, P. Abbeel, P. Sundaresan, Q. Chen, Q. Vuong, R. Rafailov, R. Tian, R. Doshi, R. Mart’in-Mart’in, R. Bajjal, R. Scalise, R. Hendrix, R. Lin, R. Qian, R. Zhang, R. Mendonca, R. Shah, R. Hoque, R. Julian, S. Bustamante, S. Kirmani, S. Levine, S. Lin, S. Moore, S. Bahl, S. Dass, S. Sonawani, S. Tulsiani, S. Song, S. Xu, S. Halder, S. Karamcheti, S. Adebola, S. Guist, S. Nasiriany, S. Schaal, S. Welker, S. Tian, S. Ramamoorthy, S. Dasari, S. Belkhale, S. Park, S. Nair, S. Mirchandani, T. Osa, T. Gupta, T. Harada, T. Matsushima, T. Xiao, T. Kollar, T. Yu, T. Ding, T. Davchev, T. Z. Zhao, T. Armstrong, T. Darrell, T. Chung, V. Jain, V. Kumar, V. Vanhoucke,

- V. Guizilini, W. Zhan, W. Zhou, W. Burgard, X. Chen, X. Chen, X. Wang, X. Zhu, X. Geng, X. Liu, X. Liangwei, X. Li, Y. Pang, Y. Lu, Y. J. Ma, Y. Kim, Y. Chebotar, Y. Zhou, Y. Zhu, Y. Wu, Y. Xu, Y. Wang, Y. Bisk, Y. Dou, Y. Cho, Y. Lee, Y. Cui, Y. Cao, Y.-H. Wu, Y. Tang, Y. Zhu, Y. Zhang, Y. Jiang, Y. Li, Y. Li, Y. Iwasawa, Y. Matsuo, Z. Ma, Z. Xu, Z. J. Cui, Z. Zhang, Z. Fu, and Z. Lin. Open X-Embodiment: Robotic learning datasets and RT-X models. <https://arxiv.org/abs/2310.08864>, 2023.
- [7] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, P. D. Fagan, J. Hejna, M. Itkina, M. Lepert, Y. J. Ma, P. T. Miller, J. Wu, S. Belkhale, S. Dass, H. Ha, A. Jain, A. Lee, Y. Lee, M. Memmel, S. Park, I. Radosavovic, K. Wang, A. Zhan, K. Black, C. Chi, K. B. Hatch, S. Lin, J. Lu, J. Mercat, A. Rehman, P. R. Sanketi, A. Sharma, C. Simpson, Q. Vuong, H. R. Walke, B. Wulfe, T. Xiao, J. H. Yang, A. Yavary, T. Z. Zhao, C. Agia, R. Baijal, M. G. Castro, D. Chen, Q. Chen, T. Chung, J. Drake, E. P. Foster, J. Gao, V. Guizilini, D. A. Herrera, M. Heo, K. Hsu, J. Hu, M. Z. Irshad, D. Jackson, C. Le, Y. Li, K. Lin, R. Lin, Z. Ma, A. Maddukuri, S. Mirchandani, D. Morton, T. Nguyen, A. O’Neill, R. Scalise, D. Seale, V. Son, S. Tian, E. Tran, A. E. Wang, Y. Wu, A. Xie, J. Yang, P. Yin, Y. Zhang, O. Bastani, G. Berseth, J. Bohg, K. Goldberg, A. Gupta, A. Gupta, D. Jayaraman, J. J. Lim, J. Malik, R. Martín-Martín, S. Ramamoorthy, D. Sadigh, S. Song, J. Wu, M. C. Yip, Y. Zhu, T. Kollar, S. Levine, and C. Finn. Droid: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024.
- [8] H. Walke, K. Black, A. Lee, M. J. Kim, M. Du, C. Zheng, T. Zhao, P. Hansen-Estruch, Q. Vuong, A. He, V. Myers, K. Fang, C. Finn, and S. Levine. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning (CoRL)*, 2023.
- [9] H. Geng, F. Wang, S. Wei, Y. Li, B. Wang, B. An, C. T. Cheng, H. Lou, P. Li, Y.-J. Wang, et al. Roboverse: Towards a unified platform, dataset and benchmark for scalable and generalizable robot learning. *arXiv preprint arXiv:2504.18904*, 2025.
- [10] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, pages 2165–2183. PMLR, 2023.
- [11] J. Zhang, X. Chen, Q. Wang, M. Li, Y. Guo, Y. Hu, J. Zhang, S. Bai, J. Lin, and J. Chen. Vlm4vla: Revisiting vision-language-models in vision-language-action models. *arXiv preprint arXiv:2601.03309*, 2026.
- [12] N. Kachaev, M. Kolosov, D. Zelezetsky, A. K. Kovalev, and A. I. Panov. Don’t blind your vla: Aligning visual representations for ood generalization. *arXiv preprint arXiv:2510.25616*, 2025.
- [13] A. J. Hancock, X. Wu, L. Zha, O. Russakovsky, and A. Majumdar. Actions as language: Fine-tuning vlms into vlas without catastrophic forgetting. *arXiv preprint arXiv:2509.22195*, 2025.
- [14] S. Belkhale, T. Ding, T. Xiao, P. Sermanet, Q. Vuong, J. Tompson, Y. Chebotar, D. Dwibedi, and D. Sadigh. Rt-h: Action hierarchies using language. *arXiv preprint arXiv:2403.01823*, 2024.
- [15] L. X. Shi, B. Ichter, M. Equi, L. Ke, K. Pertsch, Q. Vuong, J. Tanner, A. Walling, H. Wang, N. Fusai, et al. Hi robot: Open-ended instruction following with hierarchical vision-language-action models. *arXiv preprint arXiv:2502.19417*, 2025.
- [16] P. Intelligence, B. Ai, A. Amin, R. Aniceto, A. Balakrishna, G. Balke, K. Black, et al. $\pi_{0.7}$: A steerable generalist robotic foundation model with emergent capabilities. *arXiv preprint arXiv:2604.15483*, 2026.

- [17] S. Gao, J. Yang, L. Chen, K. Chitta, Y. Qiu, A. Geiger, J. Zhang, and H. Li. Vista: A generalizable driving world model with high fidelity and versatile controllability. *arXiv preprint arXiv:2405.17398*, 2024.
- [18] T. Yuan, Z. Dong, Y. Liu, and H. Zhao. Fast-wam: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026.
- [19] H. Yu, J. Zhao, Y. Liu, K. Li, C. Ma, D. Zhang, Y. Hu, G. Chen, J. Xie, J. Guo, et al. Point what you mean: Visually grounded instruction policy. *arXiv preprint arXiv:2512.18933*, 2025.
- [20] Z. Wang, Y. Chen, Y. Liu, J. Ye, P. Chen, C. Lu, S. Liu, and J. Jia. Vp-vla: Visual prompting as an interface for vision-language-action models. *arXiv preprint arXiv:2603.22003*, 2026.
- [21] R. Zheng, Y. Liang, S. Huang, J. Gao, H. Daumé III, A. Kolobov, F. Huang, and J. Yang. Tracevla: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. *arXiv preprint arXiv:2412.10345*, 2024.
- [22] H. Wu, Y. Jing, C. Cheang, G. Chen, J. Xu, X. Li, M. Liu, H. Li, and T. Kong. Unleashing large-scale video generative pre-training for visual robot manipulation, 2023.
- [23] C.-L. Cheang, G. Chen, Y. Jing, T. Kong, H. Li, Y. Li, Y. Liu, H. Wu, J. Xu, Y. Yang, H. Zhang, and M. Zhu. Gr-2: A generative video-language-action model with web-scale knowledge for robot manipulation. *arXiv preprint arXiv:2410.06158*, 2024.
- [24] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu, et al. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.
- [25] C. Cheang, S. Chen, Z. Cui, Y. Hu, L. Huang, T. Kong, H. Li, Y. Li, Y. Liu, X. Ma, et al. Gr-3 technical report. *arXiv preprint arXiv:2507.15493*, 2025.
- [26] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [27] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2024.
- [28] E. Yu et al. Dm0: An embodied-native vision-language-action model towards physical ai. *arXiv preprint arXiv:2602.14974*, 2026.
- [29] J. Chen et al. Lda-1b: Scaling latent dynamics action model via universal embodied data ingestion. *arXiv preprint arXiv:2602.12215*, 2026.
- [30] T. Lin, G. Li, Y. Zhong, Y. Zou, Y. Du, J. Liu, E. Gu, and B. Zhao. Evo-0: Vision-language-action model with implicit spatial understanding. *arXiv preprint arXiv:2507.00416*, 2025.
- [31] S. Fei, S. Wang, J. Shi, Z. Dai, J. Cai, P. Qian, L. Ji, X. He, S. Zhang, Z. Fei, J. Fu, J. Gong, and X. Qiu. Libero-plus: In-depth robustness analysis of vision-language-action models. *arXiv preprint arXiv:2510.13626*, 2025.
- [32] G. R. Team, S. Abeyruwan, J. Ainslie, J.-B. Alayrac, M. G. Arenas, T. Armstrong, A. Balakrishna, R. Baruch, M. Bauza, M. Blokzijl, et al. Gemini robotics: Bringing ai into the physical world. *arXiv preprint arXiv:2503.20020*, 2025.
- [33] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

- [34] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, et al. Sam 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.
- [35] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [36] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [37] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning, 2023.
- [38] K. Fang, F. Liu, P. Abbeel, and S. Levine. Moka: Open-world robotic manipulation through mark-based visual prompting. *Robotics: Science and Systems (RSS)*, 2024.
- [39] X. Li, L. Xu, M. Zhang, J. Liu, Y. Shen, I. Ponomarenko, J. Xu, L. Heng, S. Huang, S. Zhang, et al. Crayonrobo: Object-centric prompt-driven vision-language-action model for robotic manipulation. *arXiv preprint arXiv:2505.02166*, 2025.
- [40] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. *arXiv preprint arXiv:2409.01652*, 2024.
- [41] S. Huang, H. Chang, Y. Liu, Y. Zhu, H. Dong, P. Gao, A. Boularias, and H. Li. A3vlm: Actionable articulation-aware vision language model. *arXiv preprint arXiv:2406.07549*, 2024.
- [42] Z. Qi, W. Zhang, Y. Ding, R. Dong, X. Yu, J. Li, L. Xu, B. Li, X. He, G. Fan, et al. So-far: Language-grounded orientation bridges spatial reasoning and object manipulation. *arXiv preprint arXiv:2502.13143*, 2025.
- [43] H. Liu, S. Guo, P. Mai, J. Cao, H. Li, and J. Ma. Robodexvlm: Visual language model-enabled task planning and motion control for dexterous robot manipulation. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1381–1388. IEEE, 2025.
- [44] T. Dai, M. Han, T. Du, Z. Liu, Z. Li, S. Khan, J. Yu, and X. Chang. See, plan, rewind: Progress-aware vision-language-action models for robust robotic manipulation. *arXiv preprint arXiv:2603.09292*, 2026.
- [45] Y. Li, Z. Gong, H. Li, X. Huang, H. Kang, G. Bai, and X. Ma. Robotic visual instruction. *arXiv preprint arXiv:2505.00693*, 2025.
- [46] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. *arXiv preprint arXiv:2303.05499*, 2023.

A Additional Experimental Results

A.1 Visual Primitives Ablation

We conduct additional ablation studies to analyze the impact of different visual primitives designs on manipulation performance. Specifically, we investigate (i) primitives types and (ii) mask opacity.

All experiments are conducted using 25,282.7 seconds of training data for 10k steps, and evaluated on 50 instructions from the main evaluation benchmark.

Primitive Type. We compare four types of visual primitives: *raw* (no prompt), *point*, *box*, and *box-mask*. Results are summarized in Table 6.

Prompt Type	Instr.	Pick	Place
Raw	100	70	64
Point	100	86	74
Box	100	82	68
Box-mask	100	86	70

Table 6: **Effect of visual prompt types.**

Mask Opacity. We further analyze the effect of mask opacity in the box-mask setting. Results are summarized in Table 7.

Opacity	Instr.	Pick	Place
$\alpha = 0$	100	82	68
$\alpha = 0.7$	100	86	70
$\alpha = 0.9$	100	86	74

Table 7: **Effect of mask opacity in box-mask prompts.**

Discussion. Introducing visual prompts significantly improves performance compared to the raw setting, confirming the importance of spatial grounding. Among different prompt types, point and box-mask achieve stronger performance, while mask opacity mainly affects placement accuracy. These results suggest that precise spatial highlighting is beneficial for manipulation.

A.2 Visualization of Cross-Domain Generalization results

We provide further qualitative visualizations of our zero-shot cross-domain generalization experiments in fig. 4. As shown, the leftmost column displays a diverse set of unseen, out-of-distribution (OOD) test objects, while the subsequent columns depict the successful continuous grasping trajectories executed by AVP. While the $\pi_{0.5}$ baseline exhibits severe degradation under such distribution shifts, our method maintains remarkable robustness. By leveraging explicit visual prompts as an intermediate representation, AVP effectively decouples spatial control from specific object semantics, enabling the emergence of highly transferable, object-agnostic manipulation skills.

B Details of Action-Centric Visual-Primitive Supervision

This section provides full details of the proposed action-centric visual-primitive supervision pipeline. Our goal is to construct reliable primitive labels directly from robot interaction trajectories, avoiding the semantic ambiguity and maintenance cost of external perception-based annotation pipelines.



Figure 3: **OOD object gallery for cross-domain generalization.** We evaluate zero-shot transfer on 45 unseen objects spanning multiple categories beyond the Chinese chess training domain.

Overview. We decompose the supervision construction process into three sequential stages: (i) kinematic keyframe extraction, (ii) spatial pose estimation, and (iii) discretized primitive projection. The key principle is to determine *when* interaction occurs from robot kinematics and *where* the corresponding primitive should be spatially grounded via geometric projection.

Kinematic keyframe extraction. We first identify critical interaction keyframes by detecting substantial transitions in the gripper state. Let g_t denote the gripper state signal at time step t , such as the discrepancy between the gripper control command and the measured physical aperture. We define the interaction keyframe set as

$$T_{\text{key}} = \{ t \in [1, T] \mid |\Delta g_t| > \delta \}, \quad (9)$$

where δ is a predefined threshold for detecting significant gripper state transitions. Intuitively, these keyframes correspond to physically meaningful interaction events such as grasp initiation or release.

Spatial pose estimation. For each keyframe $t \in T_{\text{key}}$, we extract the corresponding 3D end-effector position

$$P_t = [X_t, Y_t, Z_t]^T \in \mathbb{R}^3 \quad (10)$$

from robot proprioception. This position provides a physically grounded estimate of the interaction location in the robot base coordinate frame.

Primitive projection. We then project the 3D interaction point into the image plane to obtain a 2D spatial anchor $m_t = (u_t, v_t)$. Let $K \in \mathbb{R}^{3 \times 3}$ denote the camera intrinsic matrix, and let $T_R^C \in SE(3)$ denote the extrinsic transformation from the robot base frame to the camera frame. The projected image coordinate is computed via standard perspective projection:

$$z_c \begin{bmatrix} u_t \\ v_t \\ 1 \end{bmatrix} = K T_R^C \begin{bmatrix} P_t \\ 1 \end{bmatrix}, \quad (11)$$

where z_c is the depth scaling factor in the camera coordinate system. The resulting 2D point (u_t, v_t) serves as the spatial anchor for constructing the primitive label.

Discretized primitive construction. Given the projected anchor (u_t, v_t) , we discretize its image-plane location into a finite spatial grid to obtain the supervision target for the visual-primitive decoder. Compared with primitive labels generated by generic perception models, these anchors are

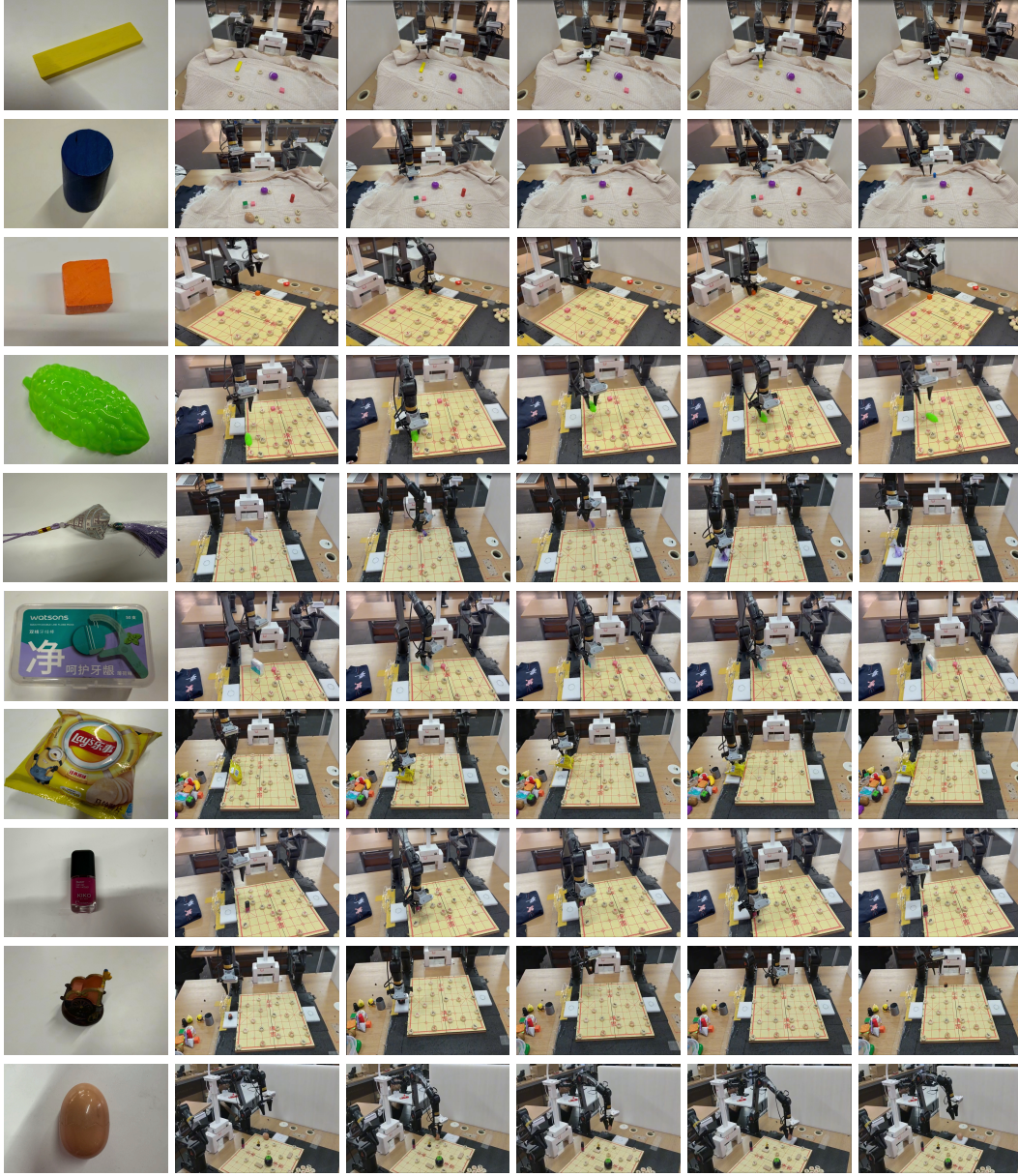


Figure 4: **Qualitative results for zero-shot cross-domain generalization.** The leftmost column displays unseen OOD test objects, and the right columns illustrate the successful grasping trajectories executed by AVP.

directly aligned with the underlying physical interaction trajectory and therefore provide more reliable spatial grounding for manipulation learning.

Discussion. This kinematics-guided supervision process offers two main advantages. First, it avoids the semantic ambiguity of external perception pipelines, especially in settings with visually similar objects, occlusion, or cluttered scenes. Second, it significantly reduces annotation cost by replacing manual or model-based primitive generation with an automated geometric pipeline. As a result, it provides a scalable and physically grounded supervision source for training the proposed visual-primitive interface.